Multi-Machine Keyboard Control

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1. Course Overview

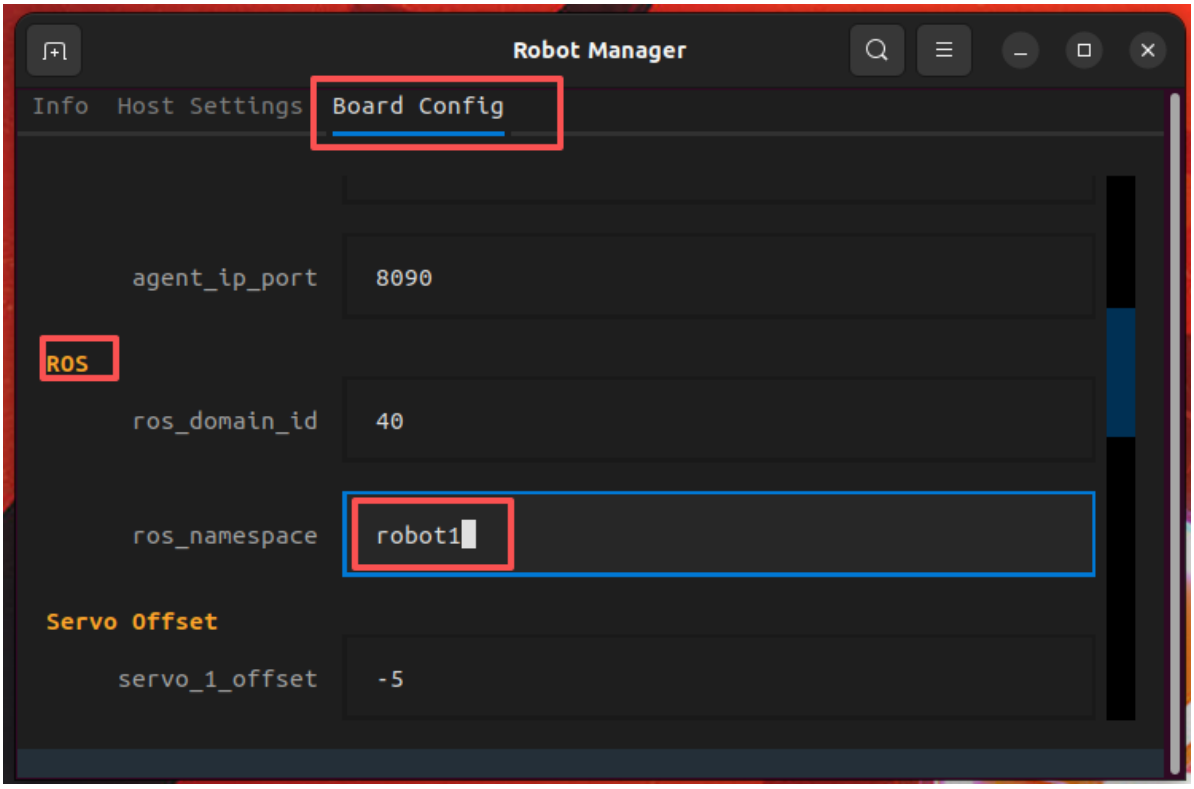
- Configure namespace for multiple robot chassis, and connect multiple MicroROSv2 robot cars via the Micro-ROS agent for synchronized control.

> [!NOTE]
>
> - This course uses 2 MicroROSv2 robot cars for demonstration.

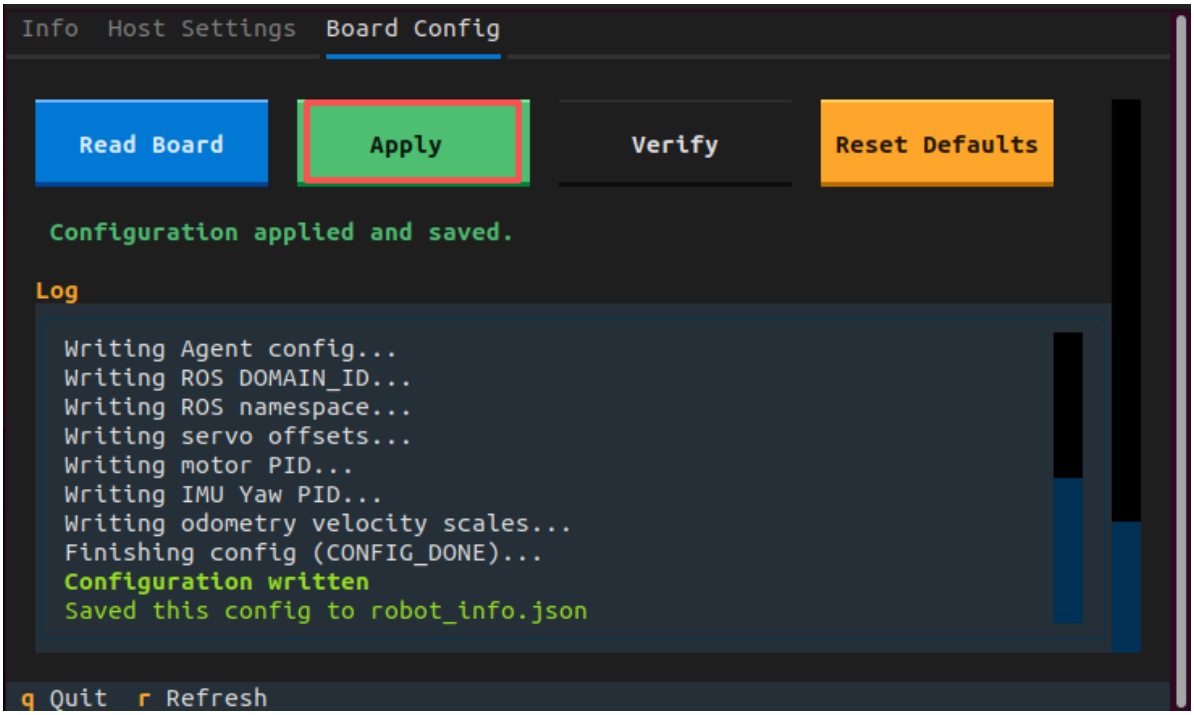
2. Preparation

2.1 Configure Robot Namespaces

- After powering on the device, use a **Type-C** cable to connect the computer to the robot car.
- Double-click the RobotManager application icon on the virtual machine desktop, select the Control page, and add the namespace of robot1 under ROS — rosnamespace.

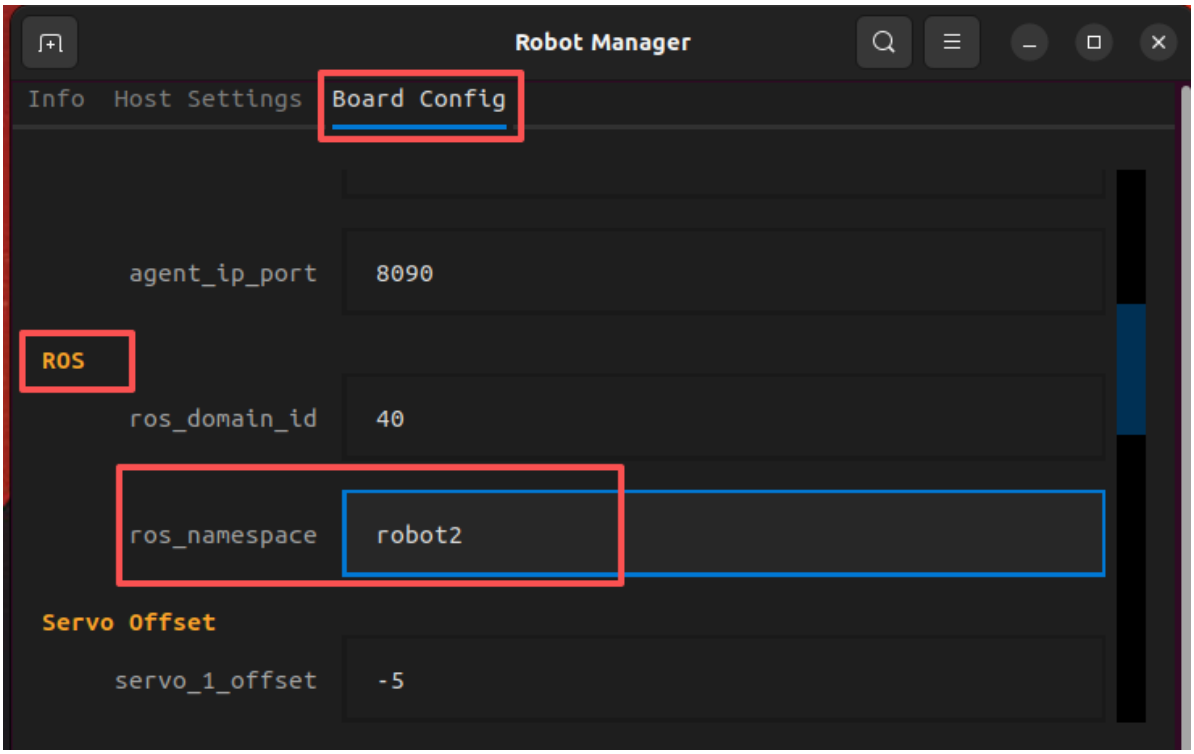


- Click Apply Configuration to write the parameters to the control board.



- **robot2**

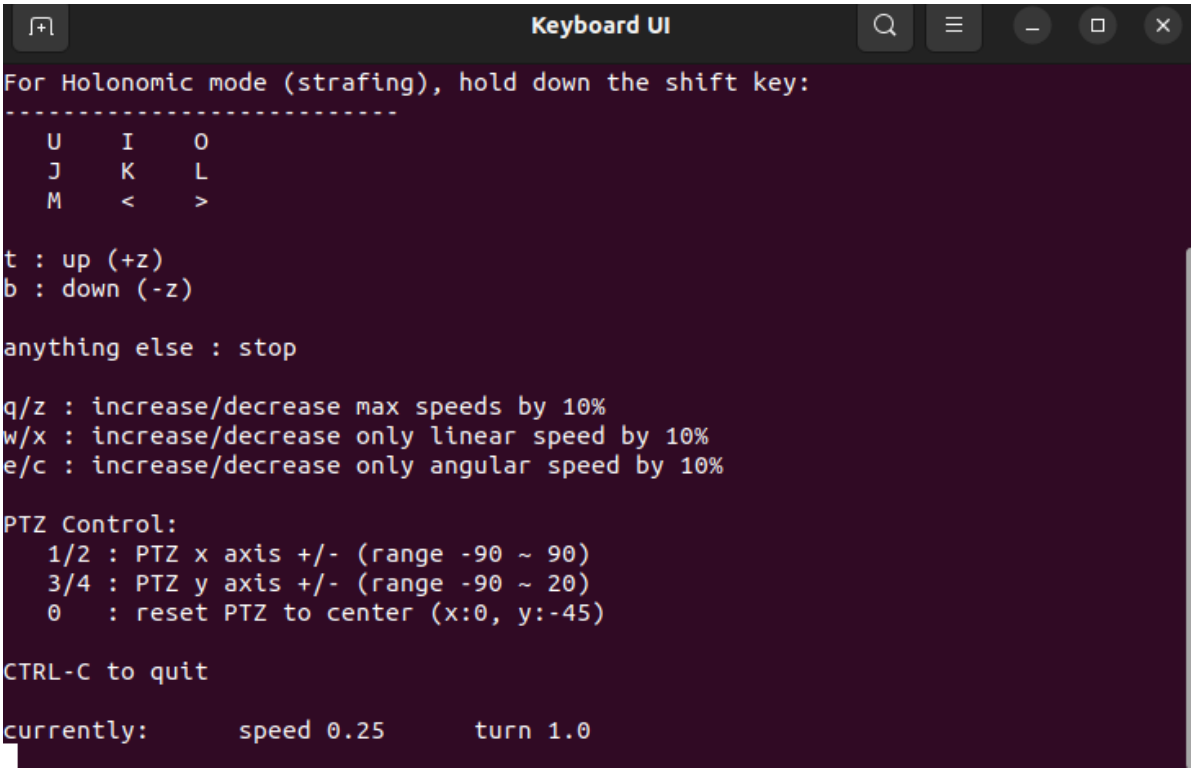
Refer to the operation steps above to change the chassis namespace on the second robot car.



3. Demonstration Case

```
ros2 launch microros_v2_bringup multi_robot_demo.launch.py demo:=keyboard
robot_names:=robot1,robot2 camera_types:=ptz,ptz
```

- Click the Keyboard UI with the mouse to perform synchronized keyboard control of the multiple robots.

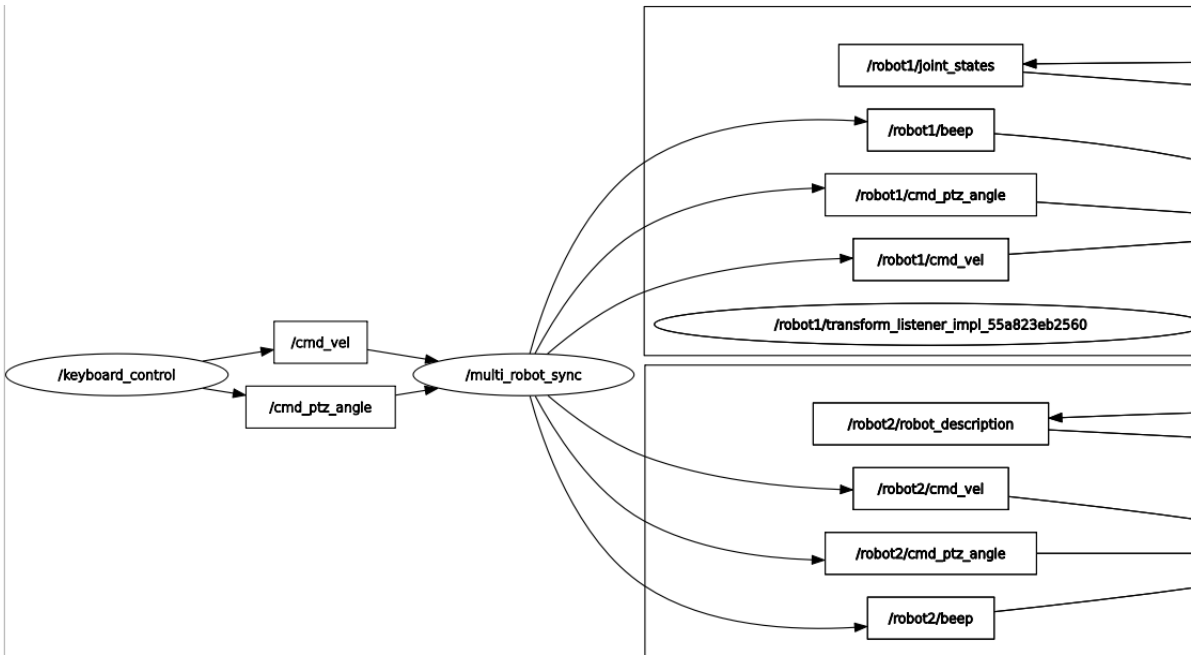


> [!TIP]
>
> - For setting up and operating multiple robots, refer to the above process. Just set a different namespace for each robot car, and make sure the parameters below actually match.
> - **robot_names**: List of robot names, separated by English commas. For example, robot1,robot2 means two MicroRosV2 robots.
> - **camera_types**: List of camera model types, separated by English commas. For example, ptz,no_ptz means the camera types of the two robots; the order matches the order of robot_names.

4. Check Node Communication

```
rqt_graph
```

- In the figure below, **multi_robot_sync** subscribes to the keyboard node's velocity control and servo PTZ control topics, and forwards the control commands to different robots.



5. Source Code Analysis

- Source code path

```
$HOME/microros_ws/src/common_demo/common_demo/multi_robot/multi_robot_sync.py
```

- Launch file path

```
$HOME/home/yahboom/microros_ws/src/microros_v2_bringup/multi_robot_demo.launch.p
y
```

- This node is a **message synchronization forwarder**. It subscribes to the common topics published by the handle, then copies and forwards the messages to the namespaced topics of each robot.
- **SYNC_TOPICS**: Defines the three topics to synchronize — `cmd_vel` (chassis velocity), `beep` (buzzer), and `cmd_ptz_angle` (PTZ angle).
- **Parameter declaration**: Receives the list of robot names through the `robot_names` parameter, with a default value of `['robot1', 'robot2']`, which can be passed in via the launch file at startup.
- **Creating publishers**: Iterates over `SYNC_TOPICS` and creates a corresponding publisher under each robot's namespace, for example `/robot1/cmd_vel`, `/robot2/cmd_vel`.
- **Creating subscribers**: Creates a non-namespaced subscriber for each topic (such as `cmd_vel`) to subscribe to the raw commands published by the handle node. Here `lambda msg, t=topic:` uses `t=topic` to correctly capture the current loop variable value in the closure.
- **sync_callback callback**: After receiving a handle message, it retrieves all publishers for that topic from `publishers_map` and forwards the message unchanged one by one, so that each robot receives the same command, achieving synchronized control.

```
SYNC_TOPICS = {
    'cmd_vel': Twist,
    'beep': UInt16,
    'cmd_ptz_angle': Vector3,
}


class MultiRobotSync(Node):
    def __init__(self):
        super().__init__('multi_robot_sync')

        self.declare_parameter('robot_names', ['robot1', 'robot2'])
        robot_names = [name.strip().strip('/') for name in
                       self.get_parameter('robot_names').value if name.strip()]
        if not robot_names:
            self.get_logger().error('robot_names is empty, nothing to sync')


        self.publishers_map = {}
        for topic, msg_type in SYNC_TOPICS.items():
            self.publishers_map[topic] = [
                self.create_publisher(msg_type, f'/{name}/{topic}', 10)
                for name in robot_names
            ]
            self.create_subscription(
                msg_type, topic,
                lambda msg, t=topic: self.sync_callback(t, msg), 10)

        self.get_logger().info(
            f'multi_robot_sync started, robots: {robot_names}, '
            f'topics: {list(SYNC_TOPICS.keys())}')

    def sync_callback(self, topic, msg):
        for pub in self.publishers_map[topic]:
            pub.publish(msg)
```